

Review article

Parkinson's disease and skin

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ABSTRACT

Parkinson's disease is associated with a variety of dermatologic disorders and the study of skin may provide insights into pathophysiological mechanisms underlying this common neurodegenerative disorder. Skin disorders in patients with Parkinson's disease can be divided into two major groups: 1) non-iatrogenic disorders, including melanoma, seborrheic dermatitis, sweating disorders, bullous pemphigoid, and rosacea, and 2) iatrogenic disorders related either to systemic side effects of antiparkinsonian medications or to the delivery system of antiparkinsonian therapy, including primarily carbidopa/levodopa, rotigotine and other dopamine agonists, amantadine, catechol-O-methyl transferase inhibitors, subcutaneous apomorphine, levodopa/carbidopa intestinal gel, and deep brain stimulation. Recent advances in our understanding of the role of α -synuclein in peripheral tissues, including the skin, and research based on induced pluripotent stem cells derived from skin fibroblasts have made skin an important target for the study of Parkinson's disease pathogenesis, drug discovery, novel stem cell therapies, and diagnostics.

1. Introduction

Parkinson's disease (PD) is the most common type of parkinsonism and the second most common progressive neurodegenerative disorder with a 1–2% lifetime risk [1,2]. It is chiefly characterized by motor symptoms (e.g. bradykinesia, tremor, rigidity, and postural instability) and non-motor symptoms (e.g. cognitive decline, depression, anxiety, overactive bladder, constipation, sialorrhea, and skin disorders). Non-motor symptoms may precede the diagnosis of PD, sometimes by several years or decades, and can be associated with substantial morbidity [3].

Skin disorders in PD are of particular interest for several reasons. Melanoma, seborrheic dermatitis (SD), sweating dysfunction, and bullous pemphigoid (BP) are overrepresented in PD [4–6]. Some skin disorders such as SD [7] and sweating dysfunction [8] may in part be due to peripheral autonomic dysfunction and can predate onset of PD motor symptoms. Melanoma [9] and BP [10], which seem to have a more complex relation to PD, often develop years after the diagnosis of

PD has been established. As these conditions can have implications on the patient's quality of life and prognosis, a high index of suspicion for PD in certain skin disorders and vice versa is therefore critically important. Skin conditions may also be seen as a consequence of oral, parenteral, and surgical therapies for PD. Such skin manifestations may be directly attributable to certain medications used in the treatment of PD, while in other cases complications related to drug delivery or therapeutic procedures may be the cause (e.g. transdermal patches, subcutaneous injection, percutaneous endoscopic gastrostomy for levodopa/carbidopa intestinal gel, and implantation of hardware in deep brain stimulation [DBS]). α -synuclein (α -syn), which plays a central role in PD pathogenesis [11], has been demonstrated over the past few decades to be present in multiple peripheral tissues, including skin [12, 13]. This small protein (140 amino acids), involved in a variety of cellular mechanisms, undergoes misfolding and aggregation in the brain as well as in the skin [14], and may contribute to some non-motor symptoms [15]. In recent years, skin biopsy has emerged as a potential biomarker and promising tool for diagnosis of α -synucleinopathies

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[16]. Furthermore, as discussed later, skin may even have therapeutic implications as a source of stem cells [17,18].

In this article we aim to provide an updated review of skin disorders associated with PD and skin-related complications of anti-parkinsonian therapy (Tables 1 and 2). We also discuss the role of skin in pathogenic, diagnostic, and therapeutic considerations in PD.

2. Methods

We conducted a review of English-language publications indexed on PubMed until June 2020 with the following search terms: Parkinson's disease OR parkinsonism AND each of the following search terms (or combinations thereof): skin, diagnosis, biopsy, sensory, autonomic, alpha synuclein, dermatitis, neuropathy, seborrhea, seborrheic dermatitis, sweat, hyperhidrosis, hypohidrosis, pemphigoid, rosacea, melanoma, cancer, nodules, treatment, levodopa, dopamine agonist, ropinirole, rotigotine, pramipexole, apomorphine, anticholinergics, monoamine oxidase, selegiline, rasagiline, safinamide, catechol-O-methyltransferase, entacapone, tolcapone, opicapone, amantadine, livedo reticularis, istradefylline, surgery, deep brain stimulation, DBS, Duopa, Duodopa, levodopa-carbidopa intestinal gel, and LCIg. The primary focus was on articles relevant to clinical aspects of skin disorders in PD. The reference lists from the articles were also reviewed.

3. Skin disorders in Parkinson's disease

3.1. Melanoma

Melanomas are malignant, pigmented tumors that arise from melanocytes typically in the skin (cutaneous melanoma) and occasionally from non-dermal melanocytes (Fig. 1). Their diagnosis can be suspected based upon visual inspection as they are characterized by asymmetry, border irregularity, color variability, large diameter, and evolution over time (the "ABCDE's"). Several types of melanomas are recognized,

including those that arise from pre-existing moles (superficial spreading melanoma), de novo (nodular melanoma), from precursor lentigo maligna (lentigo maligna melanoma), or present under nail beds or on the palms or soles (acral lentiginous melanoma). The relationship between melanoma and PD has been studied for a long time as several epidemiological studies have demonstrated increased risk of melanoma in patients with PD although the risk of other cancers is lower than expected [19–25], even after correcting for confounders known to reduce the risk of PD, such as smoking [26]. Mahajan and colleagues recently conducted a single-center, nested case-control study and found that cancer, especially skin cancer, was associated with delayed onset of PD, although the clinical progression was unaffected [27]. The odds ratio (OR; 95% confidence interval [CI]) for development of melanoma was higher after (2.43; 1.77–3.32) but not before (1.09; 0.78–1.54) diagnosis of PD in North American and European populations with no sex differences noted, according to a recent meta-analysis [28]. Similarly increased risk of melanoma after PD diagnosis was also recently noted in a large cohort of PD patients in Korea [24]. Conversely, a recent study found that patients with PD had a 3.8-fold (95% CI, 2.1–6.8) higher risk of having pre-existing melanoma [25]. In the same study, a co-diagnosis of PD seemed to confer a protective effect against death from metastatic melanoma. A positive family history of melanoma has also been shown to increase the risk of PD [29]. Not all studies support the association between PD and melanoma as noted in the meta-analyses; in a large study from Israel including more than 1.2 million members of a health organization followed from 2000 to 2015, there was no increased risk of melanoma after a diagnosis of PD, but the study suggested a higher risk of basal cell carcinoma [30]. A higher risk of non-melanoma skin cancers in PD versus controls [28] and in LRRK2-PD versus idiopathic PD [31] has also been suggested.

The association between melanoma and PD has been studied and reviewed extensively, but the nature of the relationship is still not well understood. Some studies have pointed to potential shared risk factors (higher age, male sex, Caucasian race, red hair, fair skin, and history of

Table 1
Non-iatrogenic skin disorders in Parkinson's disease.

Skin disorder	Description	Treatment/Recommendations
<i>Melanoma</i>	Skin lesions with variable color, irregular shape, and flat or raised borders, sometimes arising from pre-existing moles or de novo.	Surgical therapies, adjuvant radiation therapy, immunotherapy, and targeted therapies.
<i>Seborrheic dermatitis</i>	Sharply demarcated erythematous lesions with scaling, itching, and burning in sebaceous areas (eyebrows, nasolabial folds, ear, hairline, chest).	<u>Topical therapies:</u> Keratolytics, antifungals (e.g. ketoconazole shampoo), steroids, and topical immune suppressants. <u>Systemic therapy:</u> Antifungals, metronidazole, and antimicrobial peptides.
<i>Sweating disorders</i>	Hyperhidrosis or hypohidrosis	<u>Hyperhidrosis associated with:</u> -medication <i>OFF</i> state: increase or provide more continuous dopaminergic therapy, DBS. -dyskinesias ^a : reduce or fractionate dopaminergic therapy, start antidyskinetic therapy (amantadine). -focal symptoms: intradermal botulinum toxin injections. -STN DBS ^a : adjust stimulation parameters (dorsal to ventral stimulation for STN). <u>Hypohidrosis:</u> Avoidance of excessive heat exposure due to reduced evaporative cooling.
<i>Bullous pemphigoid</i>	Pruritic erythema and tense subepidermal blisters in flexural aspect of limbs and abdomen. (Some patients may never develop bullae.)	<u>Localized, mild/moderate disease:</u> Topical corticosteroids, oral doxycycline. <u>Severe forms:</u> Systemic corticosteroids with or without adjunctive immune suppressing or modulating therapies.
<i>Rosacea</i>	Fixed centrofacial erythema and phymatous changes, major features (flushing, papules and pustules, telangiectasia, ocular manifestations) and secondary features (burning sensation, stinging sensation, edema, dryness, other ocular manifestations).	Topical therapies, devices and surgical interventions, and oral therapies targeted against individual disease characteristics.
<i>Perioral dermatitis</i>	Perioral skin irritation associated with sialorrhea	Oral therapies (e.g. glycopyrrolate) or salivary gland injection with botulinum toxin.
<i>Neuropathy</i>	Sensory disturbance, trophic changes	Pain: alpha-2-delta ligands (e.g. gabapentin and pregabalin).

DBS = deep brain stimulation, STN = subthalamic nucleus.

^a Iatrogenic complications.

Table 2

Iatrogenic complications of Parkinson's disease therapy.

Therapy	Potential complications	Treatment/Recommendations
<i>Levodopa-carbidopa intestinal gel</i>	Wound infection/erythema at stoma site	<u>Prevention:</u> Remove dressing before cleaning. Clean with soap and water (avoid alcohol or iodine). Mobilize PEG-J 2–3 cm in and out of stoma once healed. <u>Treatment:</u> Systemic antibiotics, possible PEG-J removal. <u>Treatment:</u> Topical silver nitrate.
<i>Levodopa-carbidopa subcutaneous infusion</i>	Granulation tissue Stoma leakage Subcutaneous nodules	<u>Management:</u> Avoid topical creams/ointments. Apply dressing to keep stoma dry. <u>Prevention:</u> Rotation of infusion sites. Other potential preventative strategy under investigation. <u>Treatment:</u> Resolution 2–4 weeks after drug withdrawal.
<i>Amantadine</i>	Livedo reticularis (skin pallor surrounded by a reticular pattern of venous dilation, skin ulceration in rare cases), lower limb edema	<u>Treatment:</u> Resolution 2–4 weeks after drug withdrawal.
<i>Non-ergoline dopamine agonists</i>	Lower limb edema	<u>Treatment:</u> Resolution with drug discontinuation.
<i>Rotigotine transdermal patch</i>	Localized skin reaction (erythema, edema and or pruritus)	<u>Prevention:</u> Rotate application sites. Apply patch to clean and dry skin avoiding ointments, lotions or powders prior to application. Clean area with soap and water after patch removal. <u>Treatment:</u> Topical steroids.
<i>Apomorphine subcutaneous injection/infusion</i> <i>Sublingual apomorphine</i>	Skin nodules, infusion site erythema (pathology consistent with panniculitis)	<u>Prevention:</u> Rotation of infusion sites. Use Teflon® needles. Apply needle at steep delivery angle (45–90 degrees to the skin). Use lower concentration of apomorphine solution. <u>Treatment:</u> Massage, silicone dressing, ultrasound management.
<i>Deep brain stimulation</i>	Oropharyngeal erythema Hardware infection Skin erosion	<u>Treatment:</u> Resolution with drug discontinuation. <u>Prevention:</u> Use curved scalp incisions during placement. Avoid externalization of hardware following surgery. <u>Treatment:</u> Wound debridement, hardware removal, IV antibiotics. <u>Prevention:</u> Use of smaller hardware with lower profile, recessing hardware into drilled bone, using curved scalp incisions, implanting hardware under muscle fascia when possible. <u>Treatment:</u> Surgical revision, wound debridement, hardware removal or re-implantation

PEG-J = percutaneous endoscopic gastrostomy-jejunostomy

pesticide exposure), embryological origin of melanocytes and neurons (both arise from the neural crest), metabolic pathways, genetics, and changes in melanin synthesis which have been suggested to explain the connection [6,9,32–35]. The preponderance of evidence supporting an important link between PD and melanoma has led some organizations to recommend visiting a dermatologist at least once a year for a skin check [36]. This recommendation may be especially relevant in PD patients

**Fig. 1.** Large nodular melanoma, partially amelanocytic.

who have a particularly high risk for developing melanoma such as Caucasian individuals with frequent and intense sun exposure, the greatest risk factor for melanoma.

Levodopa, an intermediate in the biosynthetic pathway of both dopamine and melanin [37], is the mainstay of treatment of PD motor symptoms and has been hypothesized to represent the link between PD and melanoma [38,39]. However, thorough studies have not substantiated these concerns [40–42]. In a study by Fiala and colleagues, 54 patients were diagnosed with both PD and melanoma, of which 31% developed melanoma prior to even starting levodopa [40]. Another 63% took levodopa for an average (range) of 60 months (21 days–204 months) prior to diagnosis of melanoma, a duration that has been criticized as far too short as the delay between exposure to a carcinogen and development of melanoma is usually thought to be decades, not months or years [43,44]. Furthermore, there are several reports of melanoma patients who were safely treated with levodopa [40]. In a study based on data from the DATATOP trial, an association between levodopa and melanoma was also not found [41]. A large, Danish population-based study similarly found no correlation between melanoma and the estimated cumulative levodopa dose in PD [42]. Considering these findings, there does not appear to be a compelling reason to withhold levodopa in patients with melanoma or a history thereof.

As far as genetic associations are concerned, multiple genes have been implicated in both PD and melanoma [9,33]. Inzelberg and colleagues cross-referenced somatic mutations in metastatic melanoma and lung cancer (adenocarcinoma and squamous cell carcinoma) with 15 *PARK* genes using whole-exome sequencing [45]. Compared to lung cancer, significantly more patients with melanoma carried at least one (48%) or multiple (25%) *PARK* gene mutations, underscoring the genetic association between melanoma and PD. Two other studies implicated somatic *Parkin* mutations in melanoma predisposition and progression [46,47]. Recently, Dube and colleagues leveraged summary statistic datasets from several large genome wide association studies to

demonstrate a significant genetic correlation, overlapping gene expression, and 7 potential gene mediators of the connection between PD and melanoma [48].

Systemic pigmentation may also explain part of the link between PD and melanoma. Neuromelanin in the substantia nigra (SN) is chemically similar to peripheral melanin and may serve neuroprotective effects by acting as a reservoir for dopamine metabolites, drugs, and toxins [49]. Neuromelanin is reduced in the brains of patients with PD, but it is unclear if this is secondary to neuronal loss in PD or whether neuromelanin depletion is an integral part of PD-related neuropathology and if PD may in part be caused by loss of the protective effect of neuromelanin [50]. There is some evidence, however, that neuromelanin may be toxic as once released from the dying neuron it can activate resident microglia and help propagate neurodegeneration. In skin melanocytes, α -syn expression may reduce melanin synthesis by inhibiting ultraviolet radiation-induced tyrosinase activation, increasing the risk of DNA damage due to sun exposure [51]. Alterations in the gene for the melanocortin 1 receptor (MC1R), which regulates skin pigmentation, hair color, and influences melanoma risk, has been implicated as a PD risk factor [52]. Vitamin D deficiency is commonly found in both PD and melanoma and is associated with a more aggressive disease course [9]. Cutaneous pigmentation is regulated by vitamin D through enhancement of melanogenesis [53], and Vitamin D receptor subtypes may modulate the risk of both PD and melanoma [54,55]. Transcranial sonography of the midbrain reveals an extended hyperechogenic area in the SN in more than 90% of PD patients [56]. Similar changes can be seen in 42–63% of melanoma patients [57,58] and correlate with a higher incidence of prodromal symptoms of PD, in particular asymmetric motor slowing and apathy [58]. Lighter skin tone, a melanoma risk factor, is associated with a larger hyperechogenic area in the SN compared to darker skin [59].

Treatment of melanoma involves surgical excision of the tumor and surrounding healthy tissue and occasionally sentinel node biopsy in localized disease, whereas metastatic melanoma is treated with a

combination of surgery, chemotherapy, and novel targeted therapies [60].

3.2. Seborrheic dermatitis

SD is an inflammatory skin condition of variable severity characterized by sharply demarcated erythematous lesions with scaling, itching, and burning in the sebaceous areas of scalp (70% of patients), face (88%; eyebrows, nasolabial folds, and ears), and chest (27%) [61] (Fig. 2). The pathophysiology of SD is not established, but three core requirements include sebaceous gland secretion of lipids, colonization with *Malassezia* yeast that metabolize lipids and thereby generate free fatty acids and hydrogen peroxide, and a susceptible immune system [7].

SD was first reported in parkinsonism in 1927 in a study by Krestin involving mostly patients with post-encephalitic parkinsonism [62]. It has subsequently been described in PD [63–74], drug-induced parkinsonism [75], corticobasal degeneration [76], stroke and other neurological disorders [7,77,78].

Estimates of SD prevalence in PD range from 18.6% [63] to 59% [79] compared to a point prevalence of 3.2% in the general population and 4.4% in those older than 65 years [80]. Since SD is quite common, mere co-occurrence with a neurological disorder does not necessarily imply causation. SD is more common in men with or without PD compared to women, likely due to higher androgen levels (affecting sebaceous gland activity and lipid composition) in men [7].

There is no clear consensus on the possible explanation for the higher prevalence of SD in PD [7,81,82]. It has been proposed that autonomic dysfunction lead to increased sebum production which coupled with reduced spontaneous facial movements (“masked facies” or hypomimia) in turn lead to further buildup of sebum and increase the risk of SD [7]. Others have argued that reduced movements lead to changes in hygiene that promote *Malassezia* growth as well as reduce lymphatic flow preventing T-cell mediated control of *Malassezia* and may therefore better explain the link between neurological diseases, including PD, and SD [81]. A possible role of autonomic dysfunction in sebum excretion was found in one study assessing forehead sympathetic skin responses [83], but not in another report which did not assess skin autonomic function directly, but evaluated for autonomic dysfunction based on adaptation to tilting and cardiovascular tests [68]. Tanner and colleagues found that PD risk seemed to be increased following a diagnosis of SD, suggesting that SD may represent a premotor symptom due to autonomic dysfunction [84]. Higher levels of melanocyte-stimulating hormone (MSH), a “sebothrophic” hormone, has also been found in PD patients [85]. Some studies indicate that PD patients, especially male, have higher sebum excretion rates (SER) than healthy individuals [63,68,71], although SER does not seem to correlate with the severity of parkinsonism [70].

Some [64,67,71,86], but not all [63,68,87], authors have reported reduced sebum production after initiation of levodopa, however only if the SER is elevated at baseline [71]. The potential beneficial effects of levodopa may be secondary to blockade of MSH secretion and/or improved mobility leading to increased clearance of sebum [82], although the response is variable and does not appear to correlate with dose or duration of therapy. Reduction of sebum production does improve clinical characteristics of SD [86,88]. However, higher SER is unlikely to be the main explanation for the correlation between PD and SD; after all, in healthy individuals SER decrease with advancing age [89] while the prevalence of SD increases [80] and SER is usually normal in (non-PD) SD [90]. Interestingly, it has been found that SD lesions in PD have a more than two-fold increase in yeast density and in the proportion of *Malassezia* species with high lipase activity (*globosa*) compared to SD lesions in non-PD patients, although the cause of this finding is unclear [74]. Of 1408 patients with spinal cord injury in a Korean study, only 59 (4.2%) were subsequently diagnosed with SD [91], similar to the prevalence in the general population [80], arguing



Fig. 2. Seborrheic dermatitis.

against general immobility in PD and other neurological disorders as a significant contributor to SD pathogenesis. Therefore, the link between PD and SD remains as uncertain as ever.

Treatment of SD involves application of topical keratolytics, antifungals (e.g. ketoconazole shampoo), and steroids, and occasionally other topical immune suppressants (e.g. calcineurin inhibitors) and systemic therapies (antifungals, metronidazole, and antimicrobial peptides) [91,92]. Intradermal injections of botulinum toxin have been found to effectively decrease sebum production and excretion, possibly via its blockade of cholinergic signaling, in patients with oily skin disorders such as acne and SD [93,94].

3.3. Sweating disorders

Abnormalities of sweating is a common non-motor symptom in PD and include increased (hyperhidrosis) or decreased (hypohidrosis) sweating [8,63,95–109]. Many studies have focused exclusively on excessive sweating, sometimes associated with “drenching sweats”, especially during medication *OFF* periods [107,110]. Increased sweating seemed to be more common and was reported in 9–100% (average approximately 38%) of patients compared to decreased sweating in 9–46% (average approximately 15%) of patients in those studies that reported the prevalence of both problems separately in PD populations [96,100–103,106,109,111].

Abnormal sweating in PD may have a profound negative impact on quality of life [95,111]. Increased sweating affects mainly the head, neck, and trunk [104,111] and can be asymmetric [103,111]. Compared to late-onset PD, young-onset PD is associated with higher rates of sweating disorders [105,106]. Other correlates with increased sweating include camptocormia [112] and freezing of gait [113].

In contrast to hyperhidrosis associated with PD, widespread reduction of sweating (hypo- or anhidrosis) is a pattern more typical of multiple system atrophy (MSA) [103,114,115]. The presence of other symptoms of dysautonomia, greater disease severity, and higher levodopa dose also correlate with abnormal sweating [63,95,109]. Although there is no consistent relationship between treatment of PD and sweating, increased sweating is mostly seen in the medication *OFF* state likely due to insufficient central dopaminergic stimulation or in the medication *ON* state with dyskinesias, possibly due to excess energy expenditure [105,107,111,116]. However, some groups have not found convincing links between abnormal sweating and dyskinesia duration or severity [111], nonmotor fluctuations, daily dose of levodopa, or other PD medications [104,117]. Increased sweating was noted as one of the most common nonmotor symptoms in the 2- to 10-year premotor period in the ONSET PD study [8].

The pathophysiology of sweating disorders in PD is unclear, but likely involves both central dysfunction, including of sympathetic neurons in the intermediolateral cell column of the spinal cord, and peripheral autonomic denervation [118]. It has been suggested that increased sweating in axial regions may in part be a compensatory mechanism due to reduced sympathetic function in distal extremities [104].

While there is no evidence-based recommendation for the management of sweating disorders in PD [119], increased sweating has been found to improve with increased dopaminergic therapy [120] and with more continuous drug delivery (reduction of *OFF* time), such as after addition of a catechol-O-methyltransferase inhibitor to levodopa with a peripheral decarboxylase inhibitor [121], long-term subcutaneous apomorphine [122], and intrajejunal levodopa infusion [123]. However, if increased sweating occurs in the setting of dyskinesias, medication reduction may be helpful [111].

Focal increased sweating can be treated effectively with intradermal botulinum toxin injections [124]. Subthalamic nucleus (STN) DBS is associated with improvement of abnormal sweating in several reports [100,125–129], although a few authors have found no change [130] or even stimulation-induced increase of sweating, perhaps secondary to

stimulation of hypothalamic fibers [131–134].

3.4. Bullous pemphigoid

BP is an autoimmune disorder characterized by a prodrome with pruritus, erythema, and papular or urticarial cutaneous lesions followed by a bullous stage with bullae and vesicles, typically present in the flexural aspects of the limbs and abdomen [135]. The age at onset is usually in the seventh or eighth decade of life. BP is strongly associated with neurological disorders, including PD, Alzheimer's disease, stroke, and multiple sclerosis, even after correcting for age [10,136]. Up to 20% of BP cases do not progress from the “prodromal phase” into the “bullous phase” and will therefore not develop any obvious skin blistering [137]. In one study, 50.3% of patients had at least one neurological disease prior to their BP diagnosis compared with 24.2% for controls [138]; a high index of suspicion is therefore required when evaluating a patient with a known neurological disorder and a new skin condition. Neurological disorders precede BP by an average of 6.7 years [10]. Prevalence estimates of BP in PD populations range from 2.3 to 17.9% in the many published retrospective studies [138–161]. In a systematic review, there was a significant association between BP and PD with an OR (95% CI) of 3.06 (1.97–4.77) [10]. A meta-analysis using slightly less restrictive criteria (some of the included studies lacked laboratory-supported confirmation of the BP diagnosis) showed similar results [136]. BP itself is associated with a significantly increased mortality rate, although only two studies found an association between PD and increased mortality in BP [153,156].

The pathogenic link between BP and neurological disorders, including PD, is not fully understood. BP is associated with autoantibodies against two hemidesmosomal proteins: BPAg1 (BP230) and BPAg2 (BP180) both of which include neuronal and non-neuronal isoforms [10]. Neurodegeneration with associated disruption of the blood-brain-barrier, such as occurs in PD [162], may expose immunologically protected neuronal BP antigens and lead to generation of autoantibodies that cross-react with epidermal isoforms of the BP antigens [10].

Localized and mild-moderate BP is usually treated with topical corticosteroids or oral doxycycline, and severe forms is treated with systemic corticosteroids alone or in combination with adjunctive immune suppressing or modulating therapies [163].

3.5. Rosacea

Rosacea is a common inflammatory skin disorder (prevalence of up to 10% or higher in Caucasians) characterized by the presence of a fixed centrofacial erythema and phymatous changes (skin thickening and irregular surface nodularity) and/or a range of “major” (flushing, papules and pustules, telangiectasia, and ocular manifestations), and “secondary” (burning sensation, stinging sensation, edema, dryness, and ocular manifestations) features affecting primarily the convexities of the central face [164]. Evidence for a link between rosacea and PD is so far limited to 3 main studies [63,165,166]. In a cohort of patients with PD, Fischer and colleagues in 2001 first noted a relatively high prevalence of concomitant rosacea (18.8%) and flushing (31.9%), of which the latter in some cases could have represented flushing due to autonomic dysfunction in PD and not rosacea per se [63]. In a nationwide study of the Danish population (reference population >5 million individuals), Egeberg and colleagues found that a diagnosis of rosacea conferred an almost 2-fold increased risk of new-onset PD after adjusting for multiple covariates [165]. A diagnosis of rosacea was associated with 2.4 year earlier onset of PD ($p < 0.001$) and use of tetracycline seemed to reduce the risk of PD. No significant sex differences were noted. In a subsequent US study, Mathieu and colleagues [166] replicated the findings from the Danish study.

Theories based on matrix metalloproteinases and the microbiome-gut-brain axis have been offered as possible explanations for the

connection between rosacea and PD, although the pathophysiological basis for the connection is still unknown [165]. Peripheral sensory neuropathy and immune dysregulation in PD has also been offered as a possible explanation [167]. Even so, further studies are clearly needed to firmly confirm the association between PD and rosacea. Rosacea is treated with a range of topical and systemic (non-immunosuppressive) therapies and device/surgery-based therapies [168].

3.6. Perioral dermatitis

Sialorrhea is caused by pooling of saliva in the mouth and pharynx due to less frequent swallowing and hypomimia leading to involuntary leakage of saliva through the corner of the mouth causing perioral skin irritation. The prevalence of sialorrhea in PD was 40% in a recent study and was significantly associated with reduced quality of life [169]. Perioral dermatitis can develop as a consequence of uncontrolled sialorrhea and may be debilitating in severe cases [170]. Sialorrhea can be treated by reducing saliva production with oral anticholinergics such as sublingual atropine sulfate or glycopyrrolate [171], injection of botulinum toxin, particularly rimabotulinumtoxinB [172], into salivary glands, or improvement of swallowing [173].

4. Therapy-related skin disorders

4.1. Levodopa

Carbidopa-levodopa therapy is the first line therapy for motor symptoms related to PD [174]. Use of the medication rarely leads to any skin-related manifestations and is otherwise generally well tolerated. Lower extremity edema has been noted in 1–2% of patients across clinical trials [175]. There are also reports of an allergic reaction to levodopa therapy manifesting as a generalized pruritic maculopapular rash that subsequently resolves with discontinuation of levodopa treatment, however, it is thought that this reaction is more likely attributed to yellow dye sometimes used in the medication, particularly the Sinemet 25/100 formulation, rather than levodopa [175–178]. As stated previously, there is no credible connection between levodopa use and development of melanoma.

4.1.1. Levodopa-carbidopa intestinal gel

Levodopa-carbidopa intestinal gel (LCIG) is a carboxymethylcellulose gel delivered directly into the small intestine via a percutaneous endoscopic gastrostomy (PEG-J) [179]. Medication is delivered at a constant rate by using a portable infusion system and by bypassing gastric emptying which results in more steady plasma levodopa levels that helps reduce motor fluctuations seen with oral levodopa therapy. Skin-related complications related to LCIG are primarily due to complications surrounding the PEG-J placement or ongoing maintenance. In a review of five clinical trials encompassing a total of 198 patients comparing LCIG to oral medication in advanced PD in which all patients underwent PEG-J procedure, 12.9% of all patients suffered a wound infection after PEG placement [180]. Wound infection rates were higher in the control group (17.9%) than the treatment group (8.3%) suggesting that complications were due to the apparatus for delivery rather than the medication. Other complications are well documented in a review of 7 clinical trials involving 395 patients [181]. Within this cohort, 18% of patients developed excessive granulation tissue, 16% developed erythema around the incision site, and 20% developed a non-serious post-operative wound infection. Serious complications included peritonitis in 1.8% of patients and a major post procedural infection in 0.8% of patients. Similar complication rates have been reported in other studies which have additionally reported stoma leakage and subcutaneous abscess formation as potential complications [182–185]. Adverse events related to PEG-J placement for LCIG are comparable to those reported in the literature for the general population [181]. To minimize these risks patients should be instructed on proper

care of their PEG-J. Best practice guidelines include avoiding in and out movement of the PEG-J in the first 72–96 h after placement [186]. In the first 2 weeks after placement care should include removing the stoma site dressing before clearing, cleaning with only mild soap and water, and avoiding alcohol or iodine containing products. Once the stoma has healed the PEG-J tube should be mobilized 2–3 cm in and out of the stoma to prevent the PEG-J bumper from becoming embedded. Patients should avoid rotating the tubing and make sure that they dry the stoma area after activities such as showering or swimming.

Stoma infections typically occur within 3 weeks following placement. They should be treated with systemic antibiotics as topical antibiotics are generally not helpful. If there is no response to treatment the PEG-J may need to be removed. Granulation tissue can be treated topically with silver nitrate. For stoma leakage no treatment or management is indicated. Patients should be cautioned against over treating their stoma site with tropical creams and ointments. Best practices include keeping the stoma dry and applying a dressing during the day which absorbs seepage and limits any rubbing or skin irritation.

4.1.2. Levodopa-carbidopa subcutaneous infusion

Early clinical trials have assessed the safety and efficacy of a carbidopa-levodopa prodrug delivered subcutaneously through a pump (ABV-951 and ND0612). In a study of 33 healthy volunteers who received both ABV-951 and saline continuously for 10 days via subcutaneous infusion at two separate sites, the most common skin events were erythema (43%), skin “reactions” (20%), and infusion site pain (14%) [187]. In a phase 1 study of 13 PD subjects treated with subcutaneous ABV-951, steady state levodopa concentration was associated with stable levodopa exposure and minimal fluctuation [188]. Adverse events included pain at the infusion site in 2 patients. In a follow up phase 1b study, 21 PD patients with >10 years of motor symptoms received 28 days of continuous subcutaneous infusion of ABV-951 [189]. Of these patients, one patient developed a subcutaneous abscess and cellulitis. Another patient reported an infusion site infection that led to premature discontinuation.

In an open label crossover study comparing ND0612 to Levodopa/Carbidopa Intestinal Gel (LCIG) in healthy volunteers, ND0612 was shown to have similar bioequivalence of LCIG [190]. In a phase IIB open label study of ND0612 in 214 PD patients, ND0612 was noted to cause manageable skin reactions, skin nodules, infusion site hematoma, and infusion site pain. Such adverse events led to discontinuation of ND0612 in less than a quarter of patients [191]. In a phase 2 study of 38 PD patients, 24-hour ND0612 infusion was associated with significant improvement in measures of daily living following 28 days of treatment [192]. Patients reported reduction in Unified Parkinson's Disease Rating Scale Activities of Daily Living least squares (LS) mean (95% CI) scores of 2.9 points (−5.4(−0.5), $p = 0.02$), and a reduction of Parkinson's Disease Questionnaire 39 scores of 7.51 (−12.93(−2.08), $p = 0.008$).

Two phase 3 clinical trials for subcutaneous infusion of carbidopa-levodopa prodrug, ABV-951 (AbbVie, [NCT03781167](#)) and ND0612 (Neuroderm, [NCT04006210](#)), are currently underway [193]. Given promising early studies, subcutaneous carbidopa/levodopa infusion is a viable treatment option for patients with PD and levodopa-related motor fluctuations and dyskinesia [2], but as with other subcutaneous treatment options available for PD, adverse dermatologic events, such as subcutaneous nodules (Fig. 3), may influence tolerability. Potential preventive strategies, using better and safer delivery techniques, are currently being investigated.

4.2. Amantadine

Amantadine is often used in PD for the management of levodopa-induced dyskinesia. It exerts its effects through several mechanisms including competitive N-methyl-D-aspartate (NMDA) receptor antagonism, reduced striatal acetylcholine release and modulation of dopamine synthesis and uptake [194]. One of the most well documented side

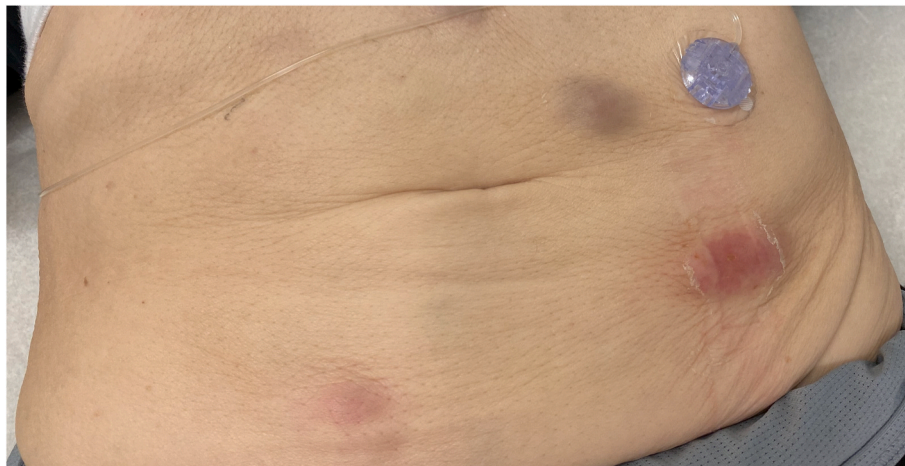


Fig. 3. Skin irritation and nodules after continuous subcutaneous infusion of carbidopa-levodopa.



Fig. 4. Livedo reticularis seen on the legs of a PD patient after starting long-acting amantadine.

effects is livedo reticularis, occurring in anywhere from 8% to 40% of patients [194–197]. Livedo reticularis is characterized by skin pallor surrounded by a reticular pattern of deoxygenated blood occurring primarily in the legs (Fig. 4), but may involve the hands as well [197]. The skin pallor is caused by arteriolar constriction leading to reduced blood flow and pallor in capillary beds. The reticulated pattern seen on the skin develops when darker deoxygenated blood accumulates in the veins that lie in between capillary beds. Proposed mechanisms of arteriole vasoconstriction include the depletion of catecholamines at peripheral nerve terminals or by a yet unknown mechanism involving

NMDA receptors in the skin [197,198]. Livedo reticularis may occur anywhere from months to years following initiation of amantadine [195]. It is typically considered a mild symptom, though patients may view it as a reason to discontinue treatment. In rare cases patients have developed skin ulceration related to livedo reticularis [199,200]. Livedo reticularis tends to resolve 2–4 weeks after discontinuation of amantadine [200].

4.3. Dopamine agonists

4.3.1. Oral dopamine agonists

Oral non-ergoline dopamine agonists used in the management of PD include pramipexole and ropinirole. Like rotigotine, these medications have high affinity for D2 and D3 receptors. Among all non-ergoline dopamine agonist, lower extremity edema is a common potential side effect, reportedly occurring in anywhere from 11% to 42% of patients [201–204]. Risk factors for developing lower extremity edema related to dopamine agonist therapy include a history of coronary artery disease and a history of diabetes mellitus [205]. There is conflicting evidence in the literature as to whether or not there is a clear dose response relationship between lower extremity edema and dosage [205,206]. Edema does not respond well to diuretic therapy, but resolves with discontinuation of the offending medication [206].

Ergoline dopamine agonists such as bromocriptine, cabergoline, pergolide and lisuride are rarely used in current clinical practice due to potential risk of valvular and lung fibrosis and safer non-ergoline options [207]. Notably, these medications have been associated with developing an erythromelalgia-like skin eruption characterized by painful erythema and edema of the feet and ankles [208,209]. In all reported cases there was resolution with discontinuation of the drug.

4.3.2. Rotigotine

Rotigotine is a non-ergoline dopamine agonist with high affinity for D2 and D3 receptors [210]. Due to extensive first pass metabolism, it has low bioavailability when taken orally, however its high lipid solubility makes it appropriate for transdermal application. Rotigotine is currently applied once every 24 hours as a transdermal patch. Steady state pharmacokinetic properties were analyzed in a review of 3 open label clinical studies in healthy patients and patients with early PD [211]. Across all patients absorption ranged from 31 to 62%, however, variability of absorption was low within patients. There was some variability in bioavailability based on application site with greater absorption in the shoulder compared to the hip and flank. Plasma concentration of rotigotine achieves a stable steady state over 24 hours of patch usage. Plasma concentrations also proportionally increased with higher doses of rotigotine and absorption does not vary if administered via a large

patch or multiple small patches.

The most common adverse event seen with rotigotine transdermal patches are local skin reactions. In a review of 6 randomized double blind placebo controlled trials (RCTs) involving a total of 2224 patients with early and advanced PD, adverse skin reactions were seen in 38–44% of patients with early PD and 15–41% of patients with advanced PD [212]. Adverse skin reactions exhibit a dose dependent relationship [213]. Typical manifestations include erythema, edema or pruritus that is limited to the patch area and may lead to discontinuation in less than 8% of patients [212,213].

Rotation of daily applications sites has been suggested to reduce the incidence of local skin reactions [213]. Patients should not use the same application site more than once every 14 days. Additional recommendations from the manufacturer include ensuring that the patch is applied to clean dry intact skin avoiding areas that could be rubbed by clothing, waistbands or skin folds. Patients should also avoid creams, lotions, ointments oils or powders over the skin area prior to application of the patch. After 24 hours the patch should be removed slowly and carefully to avoid irritation. Patients should wash previous applications sites with soap and water to remove excess drug or adhesive. Baby or mineral oil is recommended to remove excess residue. Alcohol and other solvents should be avoided. If a skin rash or irritation occurs as a result of the patch, direct sunlight on the area should be avoided until the site heals as exposure could lead to changes in skin color [213]. We have also found that application of topical steroids accelerates the resolution of the skin irritation.

A novel formulation of polymerized rotigotine called SER-214 has been developed to be delivered as once weekly injections [214]. In an open label multiple ascending dose cohort study of 19 PD patients to assess safety, patient plasma levels were shown to be highly stable with minimal fluctuation after 2 weeks of dosing. Adverse events in the study included 2 patients who reported bruising at injection sites and 1 patient who developed generalized hives and had to discontinue treatment [214].

4.3.3. Apomorphine

Apomorphine is a potent dopamine agonist that stimulates striatal D1-like and D2-like receptors and had originally been used as therapy for PD prior to the introduction of levodopa therapy [215,216]. Oral apomorphine is subject to extensive first pass metabolism by the liver, but subcutaneous administration reaches nearly 100% bioavailability and a volume of distribution and half-life comparable to if it were administered intravenously. It is indicated as a “rescue” medication taken as needed for unpredictable *OFF* symptoms of PD or as a continuous subcutaneous apomorphine infusion for PD patients who have severe motor fluctuations not controlled on oral therapy [215,216]. Skin nodules and infusion site erythema are common reactions. In one recent randomized, multicenter, double-blind, placebo-controlled trial 59% of patients in the treatment arm reported some skin changes at the injection site, 44% of patients reported skin nodules and 17% of patients reported infusion site erythema [217]. Estimates of the incidence of skin nodules in the literature vary considerably, with some reports suggesting that nodules are seen in nearly all patients receiving subcutaneous apomorphine [215,218,219]. In the majority of cases symptoms are mild and not painful. Pathological specimens of these skin nodules reveal evidence of panniculitis with fat necrosis and, in 50% of samples, large numbers of eosinophils suggesting a possible hypersensitivity reaction in addition to skin irritation as a potential explanation for the pathophysiology [220]. In rare cases necrotic skin ulcers have been reported in association with apomorphine subcutaneous infusion [221–224]. Recommendations based on expert consensus for prevention and management of skin nodules include rotating the choice of infusion sites, using Teflon® needles, using an angle of 45–90 (degrees) for delivery, using lower concentration of apomorphine solution, and maintaining skin hygiene at the infusion sites [225]. Ultrasound management has been shown to accelerate normalization of tissue in skin nodules and

improve nodule severity [219,226]. Other recommendations for treatment of skin nodules involve use of silicone gel dressings and massaging the area [219,225]. Recently, the US Food and Drug Administration approved a sublingual formulation of apomorphine as a rescue therapy of *OFF* periods in patients treated with levodopa. Although quite effective and well tolerated, the application of the thin film in the sublingual area for about 3 min per dose may be associated with oral mucosal erythema and irritation [227].

4.4. Deep brain stimulation

DBS consists of placement of intracranial electrodes within the basal ganglia, typically subthalamic nucleus, globus pallidus interna, or the ventral intermediate nucleus. The proximal end of the electrodes is connected by a subcutaneous wire to an implantable pulse generator placed subcutaneously usually in the upper chest or abdomen. It is often used in PD patients who have motor fluctuations and dyskinesia that are poorly controlled on oral medications [228]. Potential skin-related complications from DBS include hardware infection and skin erosions (Fig. 5) [229]. Hardware infection rates for DBS range from 0% to 15% in patients and an infection often precedes the development of skin erosion [229,230]. In a large systematic review of 8983 patients who underwent DBS implantation, 5.12% of patients developed hardware infection and skin erosion. The most common sites of infection were the implantable pulse generator in the chest (27%), extension cables running post auricular from chest to the DBS leads (16%), the scalp area (9%), and at the site of the burr hole (8%) [230]. Skin erosion without overlying infection was also reported in 0.8% of PD patients, most commonly at the site of the extension cable in the post auricular area (28%) followed by the IPG location (12%) [230]. Low body mass index (<24 kg/m²) is a risk factor for skin erosion [231].

Several operative practices have been associated with hardware



Fig. 5. Skin erosion over the placement site of an intracranial DBS electrode.

infection. Higher risk patients include patients who undergo straight scalp incisions compared to curved scalp incisions and patients who undergo staged procedures in which there is externalization of DBS hardware [232]. Operative techniques that isolate the lead from the skin surface at the incision sites can reduce the risk of developing hardware infection [233]. Hardware infections are treated based on the site and area involved. Appropriate management involves aggressive treatment that includes wound debridement, partial or total DBS hardware removal and/or tailored IV antibiotics [230,234]. The most common causative organism is staphylococcus aureus (20%) [230]. Skin erosion is managed by surgical revisions that may include placement of a musculocutaneous flap, wound debridement, and hardware removal or re-implantation. Recommendations to minimize the risk of skin erosion include using smaller hardware with a lower profile, recessing larger hardware components into drilled bone, using curvilinear cranial incisions as opposed to straight incisions, applying anchoring sutures on deep fascial layers rather than dermis, and, in thin patients, implanting the pulse generator under or directly onto or under the pectoralis fascia rather than into the subcutaneous fat [235,236].

5. Skin biopsy

5.1. Diagnosis of Parkinson's disease

The neuropathological hallmark of PD consists of marked loss of pigmented dopaminergic neurons in the substantia nigra and abnormal filamentous inclusions of α -syn as the major component of Lewy pathology (LP), including Lewy bodies (neuronal cytoplasmic inclusions) and Lewy neurites (in neuronal processes) [2,237]. However, the diagnosis of PD in life is based on clinical criteria [238,239] with a diagnostic accuracy of about 70–80% compared with the neuropathological gold standard even when the diagnosis is made by a movement disorders expert [240]. Such findings highlight the need for a reliable and accurate biomarker in order to enhance diagnostic confidence, especially in early PD where accuracy might be particularly low [241]. Skin biopsy may help separate α -synucleinopathies from other proteinopathies such as tauopathies (e.g. progressive supranuclear palsy) [242–244].

Abnormal α -syn aggregates have been demonstrated in various peripheral tissues, including the skin, salivary glands, adrenal glands, heart, retina, olfactory mucosa, and gastrointestinal tract [12,13]. Peripheral α -syn pathology may precede widespread central disease and thus potentially account for prodromal symptoms of PD (dysautonomia, including hypo- or hyperhidrosis, constipation, orthostatic hypotension, urinary and erectile dysfunction, anosmia etc.) by disrupting axonal function and synaptic transmission.

Native α -syn (nSyn) is expressed both in the central nervous system (CNS) and peripheral nervous system (PNS), including in dermal nerve fibers in healthy individuals and in patients with PD. Phosphorylated α -syn (pSyn) is pathological and the main constituent of LP [245], although the role of phosphorylation in the biology and pathophysiology of the protein remains controversial [246]. A recent study demonstrated that abnormal dermal α -syn aggregates are similar to midbrain α -syn in PD patients, suggesting that demonstration and quantification of abnormal dermal α -syn may be a suitable surrogate for identification of CNS pathology and a relevant outcome marker in clinical trials [14]. Dermal pSyn is found mainly in autonomic nerve terminals, richly innervating dermal autonomic annexes (blood vessels, pilomotor muscles, and sweat glands) [16]. Most immunohistochemistry studies have utilized two main types of antibodies: against pSyn (anti-pSyn) and against nSyn (anti-nSyn). While positive anti-nSyn staining in dermal nerve fibers can be shown in individuals with PD and in healthy controls [247], detection of abnormal (phosphorylated) α -syn aggregates with anti-pSyn is highly specific for PD [248]. Conversely, false-positive staining of pSyn does not seem to be a relevant concern in healthy individuals as pSyn is expressed at very low levels in healthy nervous tissue [245].

The diagnostic utility of skin biopsy in the diagnosis of PD has been evaluated in post-mortem studies [249–252] and several in vivo studies [14,247,248,253–264]. A recent meta-analysis by Tsukita et al. of in vivo skin biopsy in PD included 3 studies utilizing anti-nSyn and 9 studies utilizing anti-pSyn antibodies, all with control groups excluding other α -synucleinopathies (i.e. dementia with Lewy bodies [DLB], MSA, and pure autonomic failure [PAF]) [265]. Pooled sensitivity and specificity were higher for anti-pSyn (0.76 and 1.00, respectively) than for anti-nSyn (0.76 and 0.60, respectively). For anti-pSyn, higher number of biopsy sites (≥ 5 versus ≤ 4) also correlated with higher sensitivity (0.90 versus 0.67, respectively) but not specificity (1.00 versus 1.00), whereas the number of examined sections or the specific anti-pSyn antibody did not significantly alter diagnostic accuracy. However, an analysis of pSyn positivity rate in relation to both skin section thickness and number was not performed; since the pSyn positivity rate increases with higher number [259,261] and thicker sections analyzed [266], this likely influenced the observed pSyn rate. Furthermore, in contrast to the results reported in the aforementioned meta-analysis [265], the recently published multicenter Systemic Synuclein Sampling Study (S4) found a low sensitivity of just 24.1%, with an unchanged 100% specificity, for skin biopsy in the diagnosis of PD [267].

The exact skin biopsy site influences sensitivity of pSyn detection, with proximal sites (especially the posterior cervical region) demonstrating the highest sensitivity compared to distal sites in most [242,248,259,260], but not all [253], studies of PD and DLB. This may suggest that abnormal α -syn arise in autonomic ganglia close to the spine and then spread to the closest autonomic terminals first and later affects autonomic terminals in distal sites [16]. Conversely, patients with PD and orthostatic hypotension (PD-OH), DLB with OH (DLB-OH), and PAF exhibit more widespread (proximal and distal) pSyn deposition in skin autonomic nerves [242,243,268,269]. However, serial skin biopsies in a patient with early PD-OH showed centrifugal spread of pSyn from autonomic ganglia to target structures, but pSyn was first detected in the distal leg and thigh with spreading to cervical sites only detected 2 years later [270]. The strain of α -syn found in MSA is conformationally different from that in PD [271], and the accumulation of pSyn is typically seen in distal cutaneous somatic (not autonomic) nerves with relatively preserved autonomic fibers in MSA [261,268]. The skin sampling site may therefore be of little importance in PD-OH, DLB-OH, and PAF except early in the course, whereas proximal sites have higher sensitivity in PD and DLB without OH, and distal sites may have higher sensitivity in MSA.

Recently, detection of oligomeric α -syn using a proximity ligation assay was proposed to identify abnormal α -syn at a stage in PD prior to phosphorylation of α -syn aggregates [255]. In two recent studies using real-time quaking-induced conversion (RT-QuIC), first used on patients with Creutzfeldt-Jacob disease, pathological α -syn was detected in post-mortem posterior scalp skin biopsies from pathologically confirmed PD patients with 96–100% sensitivity and specificity [272,273]. This and other techniques, such as Protein Misfolding Cyclic Amplification (PMCA) [273], have the possibility of bringing us closer to finding a pathognomonic biomarker for PD.

Published studies of skin biopsy in the diagnosis of PD have been quite heterogeneous, especially related to sampling and immunohistochemical techniques. Additional large-scale, multicenter studies are needed to standardize and compare assessment techniques and to evaluate the diagnostic accuracy of skin biopsy in the diagnosis of PD before routine clinical use can be recommended.

5.2. Skin fibroblasts and therapy of Parkinson's disease

Skin fibroblasts from patients with PD have served as a source of induced pluripotent stem cells (iPSCs) with the possibility of using these cells in the treatment of PD and in studying PD pathophysiology and drug screening [274]. Although a detailed discussion about novel therapeutics based on skin biopsy is outside the scope of this review, a recent

and highly controversial single case report (“n-of-1” study) is worth mentioning. Schweitzer and colleagues described a 69-year-old man with PD who had moderate parkinsonism and limited treatment-related complications (mild motor fluctuations and no dyskinesias), suggestive of suboptimal dosing of dopaminergic medications [17]. The patient underwent implantation of patient-derived midbrain dopaminergic progenitor cells developed from skin fibroblasts into the putamina. Following implantation, the patient experienced some improvement of symptoms according to the authors. However, the report raises important clinical, scientific, and ethical concerns [18]. Despite these concerns, autologous iPSCs as a source of stem cells hold promise for future therapeutic applications.

6. Conclusion

Skin-related manifestations of PD are common and well documented in the literature. However, these disorders are often overlooked or disregarded by practitioners, including neurologists and dermatologists. Conditions such as SD or sweating dysfunction may precede the diagnosis of PD and could serve as potential diagnostic clue in early presentation of the conditions. As with other manifestations of PD, these conditions should not be dismissed as SD and sweating dysfunction may be responsive to levodopa therapy. Melanoma is seen more commonly in PD than in the general population, possibly due to shared risk factors and genetics. Iatrogenic complications related to PD therapy are common and may affect patient’s wellbeing or lead to discontinuation of therapy. These can include mild discomfort due to side effects from oral therapies (rash, peripheral edema etc), subcutaneous nodules due to infusion therapies, or skin erosions and infections related to surgical hardware. The skin also plays an important role in PD therapy as a target for both transdermal and subcutaneous application of medical therapy, as an emerging target for diagnostic skin biopsies, and as a possible source of patient-derived stem cells for novel disease-modifying therapies in PD.

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Andrew Billnitzer, MD: Drafted the manuscript for intellectual content.

Joseph Jankovic, MD: Conceptualized study, revised the manuscript for intellectual content.

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